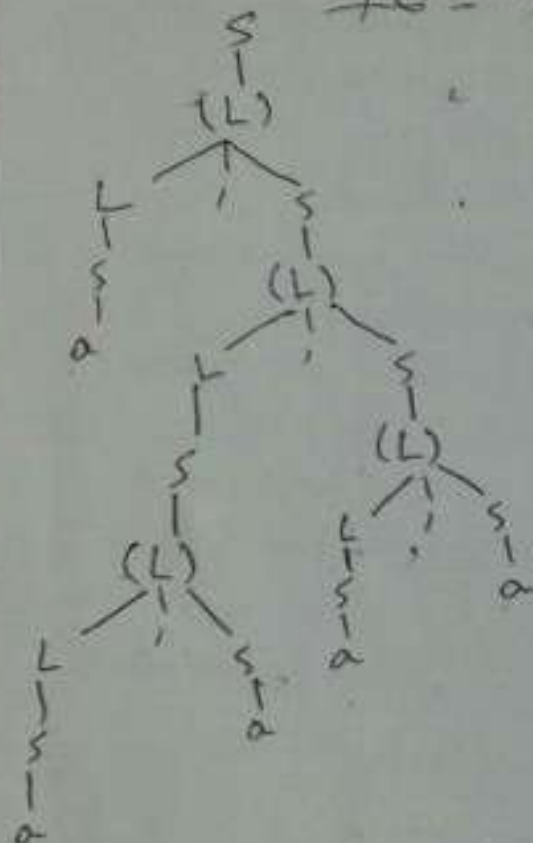
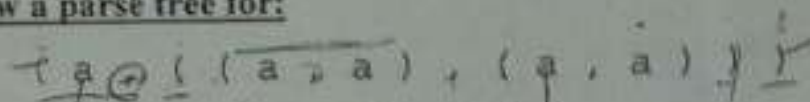


[4 Points] **Question 3:** Consider the following grammar

$$S \rightarrow (L) \mid a$$
$$L \rightarrow L, S \mid S$$

Note that the parentheses and comma are actually terminals in this language and appear in the sentences accepted by this grammar. Try to describe the language generated by this grammar. Show a parse tree for:

~~$$\begin{aligned}
 S &\rightarrow (L) \\
 &\rightarrow (L, S) \\
 &\rightarrow (S, S) \\
 &\rightarrow (a, (L)) \\
 &\rightarrow (a, ((L))) \\
 &\rightarrow (a, ((L, S))) \\
 &\rightarrow (a, ((S, S))) \\
 &\rightarrow (a, ((a
 \end{aligned}$$~~

✓ 4

$$2x > 1 - 1$$

$$\frac{2x}{2} > \frac{0}{2}$$

$$x > 0$$

[2 Points] Question 4: Write four accepted strings by the following regular expressions

2) `\d+[abc]\s?`;

11b; 55b; 55b;

1. 555 ;

----- 5b;

SSQ :

2) $\{9 \setminus w(5)\} \quad 1 \in ;$

22 01

2000

 2×112

57-0111

2

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14.5

[5 Points] **Question 1:** Determine whether the following sentences are **TRUE** or **FALSE**.
Correct the **FALSE** statements.

1. (F) Cobol language has dominated scientific computing for a long time. ✓
2. () The acronym BNF stands for Backus-Naur Form. ✓
3. (T) Prolog and Lisp languages have dominated artificial intelligence. ✓
4. () In an arithmetic expression, an operator is generated lower in the parse tree has precedence over an operator produced higher up in the tree. ✓
5. () A context-free grammar is said to be ambiguous if and only if some string in the language has more than one parse tree. ✓
6. () The cost of software maintenance depends on a number of language characteristics; primarily, reliability. ✓
7. () In BNF rules, the left hand side consists of terminal and non-terminal symbols. ✓

[4 Points] **Question 2:** Choose the correct answer for each of the following statements:

- 1) Consider this Java statement: `index = 2 * count + sum;` which of the following is NOT a token:
a. index ✓
b. Identifier
c. equal_sign ✗
d. semicolon ✗
- 2) Aliasing is harmful for:
a. Writability
b. Reliability ✓
c. Readability ✓
d. Flexibility ✓
- 3) Which produces the fastest program execution?
a. Pure interpreter
b. Compiler ✓
c. Hybrid interpreter
d. Virtual machine ✓
- 4) What primitive control statement often is used to build more complicated control statements in languages that lack them?
a. for
b. if
c. goto ✓
d. while ✓
- 5) Select the weakest precondition for each of the following statements
 $x = x + y;$
 $y = x - y;$
 $x = x - y; \{x = a \text{ and } y = b\}$
a) $\{x = a \text{ and } y = -b + x\}$ c) $\{x = b \text{ and } y = b + 9\}$
b) $\{x = b \text{ and } y = -b + 9\}$ d) none of the above

- 9) (~~T~~F) The garbage collection attempts to get back memory occupied by objects that are no longer accessible to the user program.
- 10) (T~~F~~) It is possible to have multiple variables with the same address.

[3 Points] **Question 3:** Consider the following grammar to define identifiers

$\langle \text{identifier} \rangle \rightarrow \$ \langle \text{letter} \rangle \{ \langle \text{letter} \rangle | \langle \text{digit} \rangle \}$
 $\langle \text{letter} \rangle \rightarrow a | b | c | d$
 $\langle \text{digit} \rangle \rightarrow 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 0$

Which of the following identifiers are accepted by the grammar? (Yes/No)

- | | |
|----------------------------|------------------------------|
| 1. (<u>✓</u>) \$b1234 | 4. (✓) d1234 |
| 2. (<u>✓</u>) \$abcd1234 | 5. (✓) \$L1234 |
| 3. (<u>✓</u>) \$\$d1234 | 6. (✓) \$bbcc\$ |

[2 Points] **Question 4:** Compute the weakest precondition for the following statement and postconditions (show your answer):

$$1) x = 3 * y + 6 \{ x > 0 \}$$

$$\begin{aligned}
 3 * y + 6 &> 0 \\
 3 * y &> -6 \\
 y &> -2
 \end{aligned}$$

$y > -2$

1. [2.5 Points] **Question 5:** Write EBNF grammar rules for a BEGIN-END block. It starts with the keyword **Begin** followed by one **statement** or more followed by the keyword **End**. Consider **statement** is already defined.

[2.5 Points] **Question 6:** Write a regular expression that accept octal numbers of up to 4 digits, starting with a zero digit. (for example, accept the following numbers 0256, 0755, 0232, 0110 and reject the following numbers 5598, 6552, 148000, 4a4d)

$[0][0-7]{3}$

Name: _____

ID: _____

[5 Points] **Question 1:** Choose the correct answer for each of the following statements:

- 1) Consider this Java statement: $x = 2 * y + y$; which of the following is NOT a token:
- a. ☒ X
 - b. identifier
 - c. equal_sign
 - d. semicolon

- 2) Aliasing is harmful for:
- a. Writability
 - b. ☒ Reliability
 - c. Readability
 - d. Flexibility

- 3) Which produces the fastest program execution?
- a. Pure interpreter
 - b. ☒ Compiler
 - c. Hybrid interpreter
 - d. Virtual machine

- 4) Given the following BNF grammar:

$$\langle S \rangle \rightarrow \langle A \rangle \langle A \rangle \langle S \rangle \mid \langle B \rangle$$
$$\langle A \rangle \rightarrow a$$
$$\langle B \rangle \rightarrow aab$$

Which statement is true about the above grammar?

- a. The grammar is direct and indirect recursive
 - b. The grammar is left and direct recursive
 - c. The grammar is right and indirect recursive
 - d. The grammar is right and direct recursive
- 5) What primitive control statement often is used to build more complicated control statements in languages that lack them?
- a. for
 - b. if
 - c. goto
 - d. while
- 6) Design issues of variables names includes:
- a. Case sensitivity
 - b. Name length
 - c. Special words
 - d. All of the above
- 7) The lifetime of local variable in C language is:
- a. Static
 - b. Explicit heap-dynamic
 - c. ☒ Stack-dynamic
 - d. none of the above

- 8) (T / F) The heap is a collection of storage cells which is highly organized.